

# When Does Aggregating Explanations Work?

## DermaMNIST Results

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This report consolidates the released **DermaMNIST** experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

| Protocol             | Model     | Settings                     |
|----------------------|-----------|------------------------------|
| Paper protocol       | ResNet-18 | NAIVE, IND, NOISE-S, NOISE-K |
| Paper protocol       | ViT-B/16  | NAIVE, IND, NOISE-S, NOISE-K |
| Full-matrix protocol | ViT-B/16  | NAIVE, IND, NOISE-S, NOISE-K |

Table 1: NAIVE results for **DermaMNIST** with **ResNet-18** (Paper protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains  $n = 2005$  samples.

| Method           | Clean quality |                        |              |                        | Gaussian ( $g$ )     |                              |                      |                              | Salt-and-pepper ( $p$ ) |                              |                      |                              | Speckle ( $s$ )      |                              |                      |                              | Adversarial ( $a$ )  |                              |                      |                              |
|------------------|---------------|------------------------|--------------|------------------------|----------------------|------------------------------|----------------------|------------------------------|-------------------------|------------------------------|----------------------|------------------------------|----------------------|------------------------------|----------------------|------------------------------|----------------------|------------------------------|----------------------|------------------------------|
|                  | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | $R_F^g$ $\downarrow$ | $R_{\bar{F}}^g$ $\downarrow$ | $R_C^g$ $\downarrow$ | $R_{\bar{C}}^g$ $\downarrow$ | $R_F^p$ $\downarrow$    | $R_{\bar{F}}^p$ $\downarrow$ | $R_C^p$ $\downarrow$ | $R_{\bar{C}}^p$ $\downarrow$ | $R_F^s$ $\downarrow$ | $R_{\bar{F}}^s$ $\downarrow$ | $R_C^s$ $\downarrow$ | $R_{\bar{C}}^s$ $\downarrow$ | $R_F^a$ $\downarrow$ | $R_{\bar{F}}^a$ $\downarrow$ | $R_C^a$ $\downarrow$ | $R_{\bar{C}}^a$ $\downarrow$ |
| Best Individual  | 0.077         | 0.298                  | 0.315        | 0.662                  | 0.000                | 0.282                        | 0.047                | 0.034                        | 0.000                   | 0.331                        | 0.073                | 0.022                        | 0.147                | 0.139                        | 0.040                | 0.008                        | 0.017                | 0.025                        | 0.152                | 0.026                        |
| Simple Averaging | 0.059         | 0.380                  | 0.281        | 0.732                  | 0.019                | 0.353                        | 0.165                | 0.111                        | 0.001                   | 0.371                        | 0.187                | 0.096                        | 0.231                | 0.251                        | 0.095                | 0.005                        | 0.041                | 0.130                        | 0.218                | 0.110                        |
| Borda            | 0.051         | 0.357                  | 0.267        | 0.723                  | 0.026                | 0.399                        | 0.149                | 0.061                        | 0.006                   | 0.447                        | 0.174                | 0.014                        | 0.201                | 0.317                        | 0.078                | 0.035                        | 0.037                | 0.067                        | 0.225                | 0.077                        |
| Kemeny–Young     | 0.053         | 0.349                  | 0.275        | 0.718                  | 0.022                | 0.415                        | 0.156                | 0.053                        | 0.002                   | 0.444                        | 0.182                | 0.005                        | 0.189                | 0.308                        | 0.058                | 0.039                        | 0.043                | 0.064                        | 0.229                | 0.078                        |
| RRF              | 0.055         | 0.354                  | 0.275        | 0.717                  | 0.022                | 0.403                        | 0.156                | 0.063                        | 0.001                   | 0.446                        | 0.181                | 0.015                        | 0.187                | 0.330                        | 0.059                | 0.039                        | 0.041                | 0.070                        | 0.227                | 0.079                        |
| Schulze          | 0.054         | 0.352                  | 0.270        | 0.718                  | 0.024                | 0.409                        | 0.150                | 0.062                        | 0.001                   | 0.462                        | 0.177                | 0.012                        | 0.186                | 0.319                        | 0.062                | 0.033                        | 0.042                | 0.069                        | 0.219                | 0.080                        |

Table 2: IND results for **DermaMNIST** with **ResNet-18** (Paper protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at  $q = 11$ . All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains  $n = 2005$  samples.

| Method           | Setting       | Clean quality |                        |              |                        | Gaussian ( $g$ )                         |                                                             |                                          |                                                             | Salt-and-pepper ( $p$ )                  |                                                             |                                          |                                                             | Speckle ( $s$ )                          |                                                             |                                          |                                                             | Adversarial ( $a$ )                      |                                                             |                                          |                                                             |
|------------------|---------------|---------------|------------------------|--------------|------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|
|                  |               | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>a</sup> $\downarrow$ |
| Best Individual  | baseline      | 0.055         | 0.338                  | 0.257        | 0.714                  | 0.025                                    | 0.419                                                       | 0.137                                    | 0.196                                                       | 0.000                                    | 0.406                                                       | 0.165                                    | 0.207                                                       | 0.209                                    | 0.202                                                       | 0.121                                    | 0.128                                                       | 0.045                                    | 0.033                                                       | 0.218                                    | 0.039                                                       |
| Simple Averaging | matched NAIVE | 0.054         | 0.519                  | 0.230        | 0.854                  | 0.027                                    | 0.386                                                       | 0.110                                    | 0.126                                                       | 0.003                                    | 0.396                                                       | 0.138                                    | 0.132                                                       | 0.245                                    | 0.184                                                       | 0.152                                    | 0.085                                                       | 0.045                                    | 0.030                                                       | 0.202                                    | 0.030                                                       |
| Simple Averaging | IND           | 0.040         | 0.501                  | 0.216        | 0.826                  | 0.040                                    | 0.363                                                       | 0.097                                    | 0.163                                                       | 0.017                                    | 0.363                                                       | 0.125                                    | 0.169                                                       | 0.253                                    | 0.176                                                       | 0.154                                    | 0.113                                                       | 0.030                                    | 0.033                                                       | 0.190                                    | 0.034                                                       |
| Borda            | matched NAIVE | 0.047         | 0.565                  | 0.217        | 0.896                  | 0.033                                    | 0.402                                                       | 0.097                                    | 0.095                                                       | 0.010                                    | 0.417                                                       | 0.126                                    | 0.099                                                       | 0.243                                    | 0.178                                                       | 0.150                                    | 0.072                                                       | 0.038                                    | 0.012                                                       | 0.193                                    | 0.013                                                       |
| Borda            | IND           | 0.054         | 0.568                  | 0.220        | 0.903                  | 0.026                                    | 0.399                                                       | 0.101                                    | 0.090                                                       | 0.004                                    | 0.417                                                       | 0.129                                    | 0.092                                                       | 0.248                                    | 0.170                                                       | 0.159                                    | 0.072                                                       | 0.046                                    | 0.010                                                       | 0.198                                    | 0.014                                                       |
| Kemeny–Young     | matched NAIVE | 0.048         | 0.578                  | 0.217        | 0.909                  | 0.032                                    | 0.410                                                       | 0.097                                    | 0.083                                                       | 0.009                                    | 0.432                                                       | 0.125                                    | 0.086                                                       | 0.241                                    | 0.189                                                       | 0.149                                    | 0.062                                                       | 0.041                                    | 0.010                                                       | 0.197                                    | 0.012                                                       |
| Kemeny–Young     | IND           | 0.057         | 0.574                  | 0.229        | 0.906                  | 0.023                                    | 0.401                                                       | 0.110                                    | 0.087                                                       | 0.001                                    | 0.424                                                       | 0.138                                    | 0.091                                                       | 0.245                                    | 0.170                                                       | 0.148                                    | 0.072                                                       | 0.048                                    | 0.017                                                       | 0.205                                    | 0.015                                                       |
| RRF              | matched NAIVE | 0.050         | 0.575                  | 0.217        | 0.905                  | 0.030                                    | 0.408                                                       | 0.097                                    | 0.088                                                       | 0.007                                    | 0.430                                                       | 0.126                                    | 0.090                                                       | 0.241                                    | 0.187                                                       | 0.149                                    | 0.064                                                       | 0.042                                    | 0.006                                                       | 0.195                                    | 0.007                                                       |
| RRF              | IND           | 0.063         | 0.574                  | 0.227        | 0.908                  | 0.018                                    | 0.401                                                       | 0.107                                    | 0.087                                                       | 0.005                                    | 0.421                                                       | 0.136                                    | 0.090                                                       | 0.240                                    | 0.168                                                       | 0.155                                    | 0.071                                                       | 0.052                                    | 0.011                                                       | 0.200                                    | 0.013                                                       |
| Schulze          | matched NAIVE | 0.051         | 0.571                  | 0.220        | 0.904                  | 0.029                                    | 0.406                                                       | 0.101                                    | 0.088                                                       | 0.005                                    | 0.425                                                       | 0.129                                    | 0.092                                                       | 0.239                                    | 0.184                                                       | 0.147                                    | 0.067                                                       | 0.043                                    | 0.013                                                       | 0.198                                    | 0.014                                                       |
| Schulze          | IND           | 0.057         | 0.571                  | 0.222        | 0.906                  | 0.023                                    | 0.401                                                       | 0.102                                    | 0.087                                                       | 0.000                                    | 0.422                                                       | 0.131                                    | 0.088                                                       | 0.246                                    | 0.171                                                       | 0.157                                    | 0.070                                                       | 0.046                                    | 0.012                                                       | 0.200                                    | 0.015                                                       |

Table 3: NOISE-S results for **DermaMNIST** with **ResNet-18** (Paper protocol), using the Spearman-based rank-noise geometry. The  $q$  column gives the selected explanation-prefix length. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

| Method           | $q$ | Clean quality |                        |              |                        | Gaussian ( $g$ )                         |                                          |                                          |                                          | Salt-and-pepper ( $p$ )                  |                                          |                                          |                                          | Speckle ( $s$ )                          |                                          |                                          |                                          | Adversarial ( $a$ )                      |                                          |                                          |                                          |
|------------------|-----|---------------|------------------------|--------------|------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
|                  |     | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ |
| Simple Averaging | 2   | 0.079         | 0.297                  | 0.314        | 0.658                  | 0.002                                    | 0.399                                    | 0.191                                    | 0.116                                    | 0.024                                    | 0.401                                    | 0.220                                    | 0.032                                    | 0.173                                    | 0.303                                    | 0.037                                    | 0.000                                    | 0.062                                    | 0.048                                    | 0.245                                    | 0.068                                    |
| Borda            | 2   | 0.081         | 0.294                  | 0.304        | 0.656                  | 0.004                                    | 0.333                                    | 0.181                                    | 0.075                                    | 0.024                                    | 0.358                                    | 0.209                                    | 0.002                                    | 0.164                                    | 0.244                                    | 0.041                                    | 0.003                                    | 0.068                                    | 0.035                                    | 0.243                                    | 0.049                                    |
| Kemeny–Young     | 2   | 0.081         | 0.294                  | 0.304        | 0.656                  | 0.004                                    | 0.333                                    | 0.181                                    | 0.075                                    | 0.024                                    | 0.358                                    | 0.209                                    | 0.002                                    | 0.164                                    | 0.244                                    | 0.041                                    | 0.003                                    | 0.068                                    | 0.035                                    | 0.243                                    | 0.049                                    |
| RRF              | 2   | 0.094         | 0.312                  | 0.316        | 0.666                  | 0.015                                    | 0.323                                    | 0.197                                    | 0.078                                    | 0.038                                    | 0.363                                    | 0.223                                    | 0.013                                    | 0.169                                    | 0.214                                    | 0.049                                    | 0.001                                    | 0.075                                    | 0.041                                    | 0.248                                    | 0.048                                    |
| Schulze          | 2   | 0.081         | 0.294                  | 0.304        | 0.656                  | 0.004                                    | 0.333                                    | 0.181                                    | 0.075                                    | 0.024                                    | 0.358                                    | 0.209                                    | 0.002                                    | 0.164                                    | 0.244                                    | 0.041                                    | 0.003                                    | 0.068                                    | 0.035                                    | 0.243                                    | 0.049                                    |

Table 4: NOISE-K results for DermaMNIST with ResNet-18 (Paper protocol), using the Kendall-based rank-noise geometry. The  $q$  column gives the selected explanation-prefix length. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

| Method           | $q$ | Clean quality |                        |              |                        | Gaussian ( $g$ )                         |                                          |                                          |                                          | Salt-and-pepper ( $p$ )                  |                                          |                                          |                                          | Speckle ( $s$ )                          |                                          |                                          |                                          | Adversarial ( $a$ )                      |                                          |                                          |                                          |
|------------------|-----|---------------|------------------------|--------------|------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
|                  |     | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ |
| Simple Averaging | 3   | 0.069         | 0.384                  | 0.288        | 0.737                  | 0.010                                    | 0.319                                    | 0.168                                    | 0.141                                    | 0.011                                    | 0.339                                    | 0.196                                    | 0.125                                    | 0.230                                    | 0.196                                    | 0.099                                    | 0.038                                    | 0.047                                    | 0.111                                    | 0.221                                    | 0.107                                    |
| Borda            | 3   | 0.085         | 0.310                  | 0.313        | 0.670                  | 0.003                                    | 0.316                                    | 0.191                                    | 0.103                                    | 0.028                                    | 0.354                                    | 0.219                                    | 0.053                                    | 0.202                                    | 0.217                                    | 0.061                                    | 0.023                                    | 0.062                                    | 0.058                                    | 0.235                                    | 0.065                                    |
| Kemeny–Young     | 3   | 0.091         | 0.314                  | 0.314        | 0.670                  | 0.011                                    | 0.300                                    | 0.193                                    | 0.087                                    | 0.034                                    | 0.353                                    | 0.220                                    | 0.034                                    | 0.190                                    | 0.196                                    | 0.056                                    | 0.025                                    | 0.074                                    | 0.043                                    | 0.242                                    | 0.052                                    |
| RRF              | 3   | 0.078         | 0.310                  | 0.304        | 0.660                  | 0.002                                    | 0.295                                    | 0.183                                    | 0.094                                    | 0.021                                    | 0.344                                    | 0.210                                    | 0.039                                    | 0.206                                    | 0.203                                    | 0.069                                    | 0.035                                    | 0.058                                    | 0.046                                    | 0.231                                    | 0.054                                    |
| Schulze          | 3   | 0.083         | 0.308                  | 0.313        | 0.667                  | 0.002                                    | 0.316                                    | 0.190                                    | 0.103                                    | 0.026                                    | 0.352                                    | 0.218                                    | 0.054                                    | 0.203                                    | 0.215                                    | 0.062                                    | 0.026                                    | 0.061                                    | 0.058                                    | 0.234                                    | 0.063                                    |

Table 5: NAIVE results for DermaMNIST with ViT-B/16 (Paper protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains  $n = 2005$  samples.

| Method           | Clean quality |                        |              |                        | Gaussian ( $g$ )   |                            |                    |                            | Salt-and-pepper ( $p$ ) |                            |                    |                            | Speckle ( $s$ )    |                            |                    |                            | Adversarial ( $a$ ) |                            |                    |                            |
|------------------|---------------|------------------------|--------------|------------------------|--------------------|----------------------------|--------------------|----------------------------|-------------------------|----------------------------|--------------------|----------------------------|--------------------|----------------------------|--------------------|----------------------------|---------------------|----------------------------|--------------------|----------------------------|
|                  | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | $R_F^g \downarrow$ | $R_{\bar{F}}^g \downarrow$ | $R_C^g \downarrow$ | $R_{\bar{C}}^g \downarrow$ | $R_F^p \downarrow$      | $R_{\bar{F}}^p \downarrow$ | $R_C^p \downarrow$ | $R_{\bar{C}}^p \downarrow$ | $R_F^s \downarrow$ | $R_{\bar{F}}^s \downarrow$ | $R_C^s \downarrow$ | $R_{\bar{C}}^s \downarrow$ | $R_F^a \downarrow$  | $R_{\bar{F}}^a \downarrow$ | $R_C^a \downarrow$ | $R_{\bar{C}}^a \downarrow$ |
| Best Individual  | 0.104         | 0.217                  | 0.163        | 0.279                  | 0.000              | 0.138                      | 0.012              | 0.136                      | 0.003                   | 0.124                      | 0.001              | 0.114                      | 0.001              | 0.089                      | 0.003              | 0.083                      | 0.000               | 0.000                      | 0.003              | 0.000                      |
| Simple Averaging | 0.115         | 0.211                  | 0.171        | 0.271                  | 0.060              | 0.133                      | 0.062              | 0.129                      | 0.044                   | 0.120                      | 0.033              | 0.107                      | 0.023              | 0.091                      | 0.022              | 0.084                      | 0.020               | 0.017                      | 0.047              | 0.015                      |
| Borda            | 0.110         | 0.211                  | 0.166        | 0.271                  | 0.051              | 0.133                      | 0.052              | 0.129                      | 0.038                   | 0.120                      | 0.029              | 0.108                      | 0.018              | 0.092                      | 0.018              | 0.083                      | 0.022               | 0.016                      | 0.046              | 0.017                      |
| Kemeny–Young     | 0.116         | 0.211                  | 0.173        | 0.272                  | 0.058              | 0.135                      | 0.057              | 0.133                      | 0.041                   | 0.122                      | 0.032              | 0.110                      | 0.021              | 0.092                      | 0.019              | 0.086                      | 0.025               | 0.019                      | 0.049              | 0.018                      |
| RRF              | 0.116         | 0.214                  | 0.173        | 0.272                  | 0.058              | 0.139                      | 0.058              | 0.134                      | 0.041                   | 0.124                      | 0.033              | 0.110                      | 0.019              | 0.095                      | 0.022              | 0.087                      | 0.022               | 0.018                      | 0.049              | 0.019                      |
| Schulze          | 0.119         | 0.210                  | 0.175        | 0.271                  | 0.060              | 0.134                      | 0.058              | 0.133                      | 0.045                   | 0.121                      | 0.033              | 0.109                      | 0.023              | 0.091                      | 0.023              | 0.084                      | 0.030               | 0.023                      | 0.052              | 0.021                      |

Table 6: IND results for DermaMNIST with ViT-B/16 (Paper protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at  $q = 11$ . All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains  $n = 2005$  samples.

| Method           | Setting       | Clean quality |                        |              |                        | Gaussian ( $g$ )                         |                                                             |                                          |                                                             | Salt-and-pepper ( $p$ )                  |                                                             |                                          |                                                             | Speckle ( $s$ )                          |                                                             |                                          |                                                             | Adversarial ( $a$ )                      |                                                             |                                          |                                                             |
|------------------|---------------|---------------|------------------------|--------------|------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|
|                  |               | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub><math>\bar{F}</math></sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ | R <sub><math>\bar{C}</math></sub> <sup>a</sup> $\downarrow$ |
| Best Individual  | baseline      | 0.075         | 0.227                  | 0.124        | 0.288                  | 0.031                                    | 0.148                                                       | 0.028                                    | 0.147                                                       | 0.018                                    | 0.133                                                       | 0.001                                    | 0.123                                                       | 0.002                                    | 0.098                                                       | 0.003                                    | 0.094                                                       | 0.004                                    | 0.009                                                       | 0.023                                    | 0.010                                                       |
| Simple Averaging | matched NAIVE | 0.076         | 0.225                  | 0.120        | 0.287                  | 0.032                                    | 0.147                                                       | 0.021                                    | 0.146                                                       | 0.024                                    | 0.133                                                       | 0.003                                    | 0.122                                                       | 0.005                                    | 0.099                                                       | 0.004                                    | 0.094                                                       | 0.012                                    | 0.014                                                       | 0.027                                    | 0.017                                                       |
| Simple Averaging | IND           | 0.078         | 0.224                  | 0.129        | 0.287                  | 0.030                                    | 0.146                                                       | 0.028                                    | 0.147                                                       | 0.024                                    | 0.133                                                       | 0.014                                    | 0.122                                                       | 0.008                                    | 0.098                                                       | 0.009                                    | 0.095                                                       | 0.006                                    | 0.014                                                       | 0.031                                    | 0.015                                                       |
| Borda            | matched NAIVE | 0.071         | 0.229                  | 0.119        | 0.291                  | 0.027                                    | 0.151                                                       | 0.022                                    | 0.149                                                       | 0.020                                    | 0.136                                                       | 0.003                                    | 0.125                                                       | 0.005                                    | 0.101                                                       | 0.001                                    | 0.097                                                       | 0.005                                    | 0.013                                                       | 0.028                                    | 0.015                                                       |
| Borda            | IND           | 0.075         | 0.226                  | 0.123        | 0.288                  | 0.030                                    | 0.148                                                       | 0.023                                    | 0.146                                                       | 0.023                                    | 0.134                                                       | 0.004                                    | 0.123                                                       | 0.005                                    | 0.098                                                       | 0.002                                    | 0.093                                                       | 0.003                                    | 0.010                                                       | 0.025                                    | 0.013                                                       |
| Kemeny–Young     | matched NAIVE | 0.074         | 0.229                  | 0.121        | 0.290                  | 0.031                                    | 0.150                                                       | 0.024                                    | 0.149                                                       | 0.021                                    | 0.136                                                       | 0.007                                    | 0.125                                                       | 0.004                                    | 0.101                                                       | 0.000                                    | 0.098                                                       | 0.006                                    | 0.012                                                       | 0.027                                    | 0.017                                                       |
| Kemeny–Young     | IND           | 0.080         | 0.226                  | 0.131        | 0.287                  | 0.035                                    | 0.147                                                       | 0.034                                    | 0.146                                                       | 0.024                                    | 0.133                                                       | 0.013                                    | 0.122                                                       | 0.010                                    | 0.100                                                       | 0.008                                    | 0.096                                                       | 0.007                                    | 0.012                                                       | 0.031                                    | 0.016                                                       |
| RRF              | matched NAIVE | 0.070         | 0.228                  | 0.119        | 0.289                  | 0.028                                    | 0.150                                                       | 0.021                                    | 0.148                                                       | 0.019                                    | 0.135                                                       | 0.004                                    | 0.124                                                       | 0.000                                    | 0.102                                                       | 0.001                                    | 0.097                                                       | 0.003                                    | 0.011                                                       | 0.027                                    | 0.016                                                       |
| RRF              | IND           | 0.078         | 0.225                  | 0.128        | 0.288                  | 0.032                                    | 0.148                                                       | 0.026                                    | 0.148                                                       | 0.021                                    | 0.133                                                       | 0.011                                    | 0.123                                                       | 0.006                                    | 0.099                                                       | 0.007                                    | 0.096                                                       | 0.007                                    | 0.012                                                       | 0.033                                    | 0.014                                                       |
| Schulze          | matched NAIVE | 0.073         | 0.228                  | 0.120        | 0.290                  | 0.030                                    | 0.149                                                       | 0.024                                    | 0.149                                                       | 0.021                                    | 0.135                                                       | 0.005                                    | 0.125                                                       | 0.004                                    | 0.101                                                       | 0.000                                    | 0.097                                                       | 0.005                                    | 0.013                                                       | 0.026                                    | 0.017                                                       |
| Schulze          | IND           | 0.083         | 0.227                  | 0.133        | 0.289                  | 0.038                                    | 0.149                                                       | 0.034                                    | 0.149                                                       | 0.027                                    | 0.135                                                       | 0.016                                    | 0.125                                                       | 0.015                                    | 0.102                                                       | 0.007                                    | 0.099                                                       | 0.013                                    | 0.011                                                       | 0.036                                    | 0.014                                                       |

Table 7: NOISE-S results for **DermaMNIST** with ViT-B/16 (Paper protocol), using the Spearman-based rank-noise geometry. The  $q$  column gives the selected explanation-prefix length. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

| Method           | $q$ | Clean quality |                        |              |                        | Gaussian ( $g$ )                         |                                          |                                          |                                          | Salt-and-pepper ( $p$ )                  |                                          |                                          |                                          | Speckle ( $s$ )                          |                                          |                                          |                                          | Adversarial ( $a$ )                      |                                          |                                          |                                          |
|------------------|-----|---------------|------------------------|--------------|------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
|                  |     | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ |
| Simple Averaging | 8   | 0.116         | 0.213                  | 0.176        | 0.272                  | 0.062                                    | 0.133                                    | 0.067                                    | 0.130                                    | 0.046                                    | 0.121                                    | 0.039                                    | 0.110                                    | 0.023                                    | 0.092                                    | 0.028                                    | 0.084                                    | 0.023                                    | 0.014                                    | 0.051                                    | 0.015                                    |
| Borda            | 8   | 0.116         | 0.208                  | 0.171        | 0.268                  | 0.059                                    | 0.130                                    | 0.058                                    | 0.127                                    | 0.042                                    | 0.117                                    | 0.029                                    | 0.106                                    | 0.023                                    | 0.093                                    | 0.025                                    | 0.081                                    | 0.020                                    | 0.020                                    | 0.042                                    | 0.020                                    |
| Kemeny–Young     | 8   | 0.118         | 0.207                  | 0.176        | 0.268                  | 0.057                                    | 0.131                                    | 0.059                                    | 0.128                                    | 0.043                                    | 0.118                                    | 0.034                                    | 0.107                                    | 0.019                                    | 0.088                                    | 0.023                                    | 0.082                                    | 0.024                                    | 0.025                                    | 0.049                                    | 0.024                                    |
| RRF              | 8   | 0.122         | 0.211                  | 0.181        | 0.271                  | 0.063                                    | 0.134                                    | 0.065                                    | 0.130                                    | 0.047                                    | 0.122                                    | 0.038                                    | 0.109                                    | 0.025                                    | 0.092                                    | 0.030                                    | 0.085                                    | 0.031                                    | 0.016                                    | 0.056                                    | 0.018                                    |
| Schulze          | 8   | 0.120         | 0.211                  | 0.180        | 0.273                  | 0.061                                    | 0.135                                    | 0.064                                    | 0.133                                    | 0.047                                    | 0.123                                    | 0.037                                    | 0.112                                    | 0.023                                    | 0.095                                    | 0.027                                    | 0.090                                    | 0.022                                    | 0.018                                    | 0.050                                    | 0.015                                    |



Table 8: NOISE-K results for **DermaMNIST** with ViT-B/16 (Paper protocol), using the Kendall-based rank-noise geometry. The  $q$  column gives the selected explanation-prefix length. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

| Method           | $q$ | Clean quality |                        |              |                        | Gaussian ( $g$ )                         |                                          |                                          |                                          | Salt-and-pepper ( $p$ )                  |                                          |                                          |                                          | Speckle ( $s$ )                          |                                          |                                          |                                          | Adversarial ( $a$ )                      |                                          |                                          |                                          |
|------------------|-----|---------------|------------------------|--------------|------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
|                  |     | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>C</sub> <sup>g</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>C</sub> <sup>p</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>C</sub> <sup>s</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>F</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ | R <sub>C</sub> <sup>a</sup> $\downarrow$ |
| Simple Averaging | 5   | 0.115         | 0.209                  | 0.177        | 0.269                  | 0.061                                    | 0.130                                    | 0.068                                    | 0.127                                    | 0.044                                    | 0.117                                    | 0.038                                    | 0.106                                    | 0.023                                    | 0.089                                    | 0.032                                    | 0.082                                    | 0.021                                    | 0.018                                    | 0.052                                    | 0.017                                    |
| Borda            | 5   | 0.107         | 0.208                  | 0.167        | 0.265                  | 0.055                                    | 0.130                                    | 0.062                                    | 0.124                                    | 0.038                                    | 0.115                                    | 0.029                                    | 0.101                                    | 0.020                                    | 0.091                                    | 0.027                                    | 0.078                                    | 0.004                                    | 0.021                                    | 0.029                                    | 0.022                                    |
| Kemeny–Young     | 5   | 0.121         | 0.207                  | 0.182        | 0.267                  | 0.063                                    | 0.132                                    | 0.067                                    | 0.128                                    | 0.048                                    | 0.117                                    | 0.044                                    | 0.105                                    | 0.024                                    | 0.089                                    | 0.032                                    | 0.082                                    | 0.024                                    | 0.019                                    | 0.049                                    | 0.023                                    |
| RRF              | 5   | 0.118         | 0.208                  | 0.177        | 0.268                  | 0.060                                    | 0.131                                    | 0.063                                    | 0.128                                    | 0.044                                    | 0.118                                    | 0.038                                    | 0.106                                    | 0.024                                    | 0.089                                    | 0.029                                    | 0.082                                    | 0.021                                    | 0.019                                    | 0.045                                    | 0.022                                    |
| Schulze          | 5   | 0.120         | 0.204                  | 0.181        | 0.264                  | 0.063                                    | 0.128                                    | 0.068                                    | 0.124                                    | 0.047                                    | 0.112                                    | 0.042                                    | 0.101                                    | 0.029                                    | 0.083                                    | 0.035                                    | 0.075                                    | 0.019                                    | 0.025                                    | 0.045                                    | 0.025                                    |

Table 9: NAIVE results for DermaMNIST with ViT-B/16 (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains  $n = 2005$  samples.

| Method           | Clean quality |                        |              |                        | Gaussian ( $g$ )                       |                                        |                                        |                                        | Salt-and-pepper ( $p$ )                |                                        |                                        |                                        | Speckle ( $s$ )                        |                                        |                                        |                                        | Adversarial ( $a$ )                    |                                        |                                        |                                        |
|------------------|---------------|------------------------|--------------|------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
|                  | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\uparrow$ | R <sub>F</sub> <sup>g</sup> $\uparrow$ | R <sub>C</sub> <sup>g</sup> $\uparrow$ | R <sub>C</sub> <sup>g</sup> $\uparrow$ | R <sub>F</sub> <sup>p</sup> $\uparrow$ | R <sub>F</sub> <sup>p</sup> $\uparrow$ | R <sub>C</sub> <sup>p</sup> $\uparrow$ | R <sub>C</sub> <sup>p</sup> $\uparrow$ | R <sub>F</sub> <sup>s</sup> $\uparrow$ | R <sub>F</sub> <sup>s</sup> $\uparrow$ | R <sub>C</sub> <sup>s</sup> $\uparrow$ | R <sub>C</sub> <sup>s</sup> $\uparrow$ | R <sub>F</sub> <sup>a</sup> $\uparrow$ | R <sub>F</sub> <sup>a</sup> $\uparrow$ | R <sub>C</sub> <sup>a</sup> $\uparrow$ | R <sub>C</sub> <sup>a</sup> $\uparrow$ |
| Best Individual  | 0.099         | 0.202                  | 0.153        | 0.254                  | -0.061                                 | 0.150                                  | -0.081                                 | 0.166                                  | -0.025                                 | 0.111                                  | -0.026                                 | 0.106                                  | -0.009                                 | 0.084                                  | -0.017                                 | 0.078                                  | 0.040                                  | -0.035                                 | 0.034                                  | -0.036                                 |
| Simple Averaging | 0.107         | 0.207                  | 0.153        | 0.260                  | -0.067                                 | 0.154                                  | -0.080                                 | 0.172                                  | -0.034                                 | 0.116                                  | -0.026                                 | 0.112                                  | -0.018                                 | 0.090                                  | -0.009                                 | 0.087                                  | -0.002                                 | -0.020                                 | -0.010                                 | -0.020                                 |
| Borda            | 0.101         | 0.208                  | 0.141        | 0.258                  | -0.062                                 | 0.155                                  | -0.070                                 | 0.171                                  | -0.027                                 | 0.117                                  | -0.008                                 | 0.109                                  | -0.016                                 | 0.093                                  | 0.001                                  | 0.083                                  | 0.014                                  | -0.018                                 | 0.015                                  | -0.019                                 |
| Kemeny–Young     | 0.105         | 0.206                  | 0.152        | 0.257                  | -0.065                                 | 0.154                                  | -0.077                                 | 0.170                                  | -0.031                                 | 0.116                                  | -0.019                                 | 0.109                                  | -0.013                                 | 0.093                                  | -0.005                                 | 0.084                                  | 0.003                                  | -0.020                                 | -0.004                                 | -0.022                                 |
| RRF              | 0.109         | 0.207                  | 0.152        | 0.260                  | -0.068                                 | 0.154                                  | -0.078                                 | 0.173                                  | -0.033                                 | 0.116                                  | -0.016                                 | 0.112                                  | -0.018                                 | 0.090                                  | -0.007                                 | 0.084                                  | 0.000                                  | -0.022                                 | -0.004                                 | -0.023                                 |
| Schulze          | 0.109         | 0.208                  | 0.151        | 0.260                  | -0.069                                 | 0.156                                  | -0.078                                 | 0.174                                  | -0.033                                 | 0.118                                  | -0.016                                 | 0.112                                  | -0.018                                 | 0.093                                  | -0.003                                 | 0.086                                  | 0.004                                  | -0.022                                 | 0.002                                  | -0.023                                 |

Table 10: IND results for **DermaMNIST** with ViT-B/16 (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at  $q = 11$ . All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains  $n = 2005$  samples.

| Method           | Setting       | Clean quality |                        |              |                        | Gaussian ( $g$ )                       |                                                           |                                        |                                                           | Salt-and-pepper ( $p$ )                |                                                           |                                        |                                                           | Speckle ( $s$ )                        |                                                           |                                        |                                                           | Adversarial ( $a$ )                    |                                                           |                                        |                                                           |
|------------------|---------------|---------------|------------------------|--------------|------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|
|                  |               | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>g</sup> $\uparrow$ | R <sub>C</sub> <sup>g</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>g</sup> $\uparrow$ | R <sub>F</sub> <sup>p</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>p</sup> $\uparrow$ | R <sub>C</sub> <sup>p</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>p</sup> $\uparrow$ | R <sub>F</sub> <sup>s</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>s</sup> $\uparrow$ | R <sub>C</sub> <sup>s</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>s</sup> $\uparrow$ | R <sub>F</sub> <sup>a</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>a</sup> $\uparrow$ | R <sub>C</sub> <sup>a</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>a</sup> $\uparrow$ |
| Best Individual  | baseline      | 0.071         | 0.226                  | 0.115        | 0.281                  | -0.043                                 | 0.172                                                     | -0.054                                 | 0.193                                                     | -0.014                                 | 0.134                                                     | 0.000                                  | 0.132                                                     | -0.005                                 | 0.103                                                     | 0.007                                  | 0.099                                                     | 0.012                                  | -0.015                                                    | -0.005                                 | -0.016                                                    |
| Simple Averaging | matched NAIVE | 0.070         | 0.223                  | 0.110        | 0.279                  | -0.041                                 | 0.170                                                     | -0.047                                 | 0.191                                                     | -0.016                                 | 0.131                                                     | 0.006                                  | 0.129                                                     | -0.004                                 | 0.101                                                     | 0.014                                  | 0.098                                                     | 0.015                                  | -0.017                                                    | 0.001                                  | -0.019                                                    |
| Simple Averaging | IND           | 0.070         | 0.222                  | 0.108        | 0.276                  | -0.040                                 | 0.170                                                     | -0.046                                 | 0.189                                                     | -0.020                                 | 0.130                                                     | 0.005                                  | 0.127                                                     | -0.002                                 | 0.100                                                     | 0.014                                  | 0.096                                                     | 0.020                                  | -0.013                                                    | 0.009                                  | -0.017                                                    |
| Borda            | matched NAIVE | 0.067         | 0.224                  | 0.106        | 0.281                  | -0.038                                 | 0.172                                                     | -0.043                                 | 0.193                                                     | -0.014                                 | 0.133                                                     | 0.008                                  | 0.132                                                     | -0.003                                 | 0.103                                                     | 0.016                                  | 0.100                                                     | 0.015                                  | -0.014                                                    | 0.002                                  | -0.015                                                    |
| Borda            | IND           | 0.069         | 0.224                  | 0.108        | 0.279                  | -0.039                                 | 0.171                                                     | -0.047                                 | 0.191                                                     | -0.019                                 | 0.133                                                     | 0.004                                  | 0.130                                                     | -0.010                                 | 0.102                                                     | 0.009                                  | 0.098                                                     | 0.013                                  | -0.014                                                    | 0.002                                  | -0.016                                                    |
| Kemeny–Young     | matched NAIVE | 0.069         | 0.226                  | 0.109        | 0.281                  | -0.041                                 | 0.174                                                     | -0.048                                 | 0.194                                                     | -0.017                                 | 0.135                                                     | 0.007                                  | 0.133                                                     | -0.005                                 | 0.103                                                     | 0.014                                  | 0.100                                                     | 0.015                                  | -0.012                                                    | 0.002                                  | -0.014                                                    |
| Kemeny–Young     | IND           | 0.069         | 0.223                  | 0.110        | 0.278                  | -0.040                                 | 0.172                                                     | -0.047                                 | 0.191                                                     | -0.017                                 | 0.133                                                     | 0.002                                  | 0.130                                                     | -0.005                                 | 0.102                                                     | 0.012                                  | 0.099                                                     | 0.018                                  | -0.011                                                    | 0.007                                  | -0.015                                                    |
| RRF              | matched NAIVE | 0.070         | 0.227                  | 0.110        | 0.283                  | -0.041                                 | 0.174                                                     | -0.048                                 | 0.195                                                     | -0.019                                 | 0.135                                                     | 0.006                                  | 0.134                                                     | -0.006                                 | 0.105                                                     | 0.011                                  | 0.102                                                     | 0.012                                  | -0.012                                                    | -0.001                                 | -0.013                                                    |
| RRF              | IND           | 0.071         | 0.222                  | 0.108        | 0.276                  | -0.040                                 | 0.170                                                     | -0.045                                 | 0.189                                                     | -0.017                                 | 0.131                                                     | 0.009                                  | 0.128                                                     | -0.009                                 | 0.100                                                     | 0.011                                  | 0.097                                                     | 0.016                                  | -0.014                                                    | 0.007                                  | -0.017                                                    |
| Schulze          | matched NAIVE | 0.071         | 0.227                  | 0.111        | 0.282                  | -0.042                                 | 0.174                                                     | -0.050                                 | 0.194                                                     | -0.018                                 | 0.136                                                     | 0.006                                  | 0.133                                                     | -0.006                                 | 0.105                                                     | 0.014                                  | 0.101                                                     | 0.012                                  | -0.012                                                    | -0.001                                 | -0.015                                                    |
| Schulze          | IND           | 0.069         | 0.223                  | 0.111        | 0.277                  | -0.040                                 | 0.171                                                     | -0.047                                 | 0.190                                                     | -0.014                                 | 0.132                                                     | 0.004                                  | 0.128                                                     | -0.006                                 | 0.104                                                     | 0.011                                  | 0.100                                                     | 0.018                                  | -0.014                                                    | 0.005                                  | -0.019                                                    |

Table 11: NOISE-S results for **DermaMNIST** with ViT-B/16 (Full-matrix protocol), using the Spearman-based rank-noise geometry. The  $q$  column gives the selected explanation-prefix length. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

| Method           | $q$ | Clean quality |                        |              |                        | Gaussian ( $g$ )                       |                                                           |                                        |                                                           | Salt-and-pepper ( $p$ )                |                                                           |                                        |                                                           | Speckle ( $s$ )                        |                                                           |                                        |                                                           | Adversarial ( $a$ )                    |                                                           |                                        |                                                           |
|------------------|-----|---------------|------------------------|--------------|------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|
|                  |     | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>g</sup> $\uparrow$ | R <sub>C</sub> <sup>g</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>g</sup> $\uparrow$ | R <sub>F</sub> <sup>p</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>p</sup> $\uparrow$ | R <sub>C</sub> <sup>p</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>p</sup> $\uparrow$ | R <sub>F</sub> <sup>s</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>s</sup> $\uparrow$ | R <sub>C</sub> <sup>s</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>s</sup> $\uparrow$ | R <sub>F</sub> <sup>a</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>a</sup> $\uparrow$ | R <sub>C</sub> <sup>a</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>a</sup> $\uparrow$ |
| Simple Averaging | 10  | 0.109         | 0.206                  | 0.156        | 0.259                  | -0.069                                 | 0.154                                                     | -0.083                                 | 0.171                                                     | -0.035                                 | 0.116                                                     | -0.028                                 | 0.110                                                     | -0.020                                 | 0.088                                                     | -0.011                                 | 0.084                                                     | 0.000                                  | -0.020                                                    | -0.009                                 | -0.021                                                    |
| Borda            | 10  | 0.103         | 0.205                  | 0.144        | 0.258                  | -0.065                                 | 0.152                                                     | -0.073                                 | 0.170                                                     | -0.031                                 | 0.114                                                     | -0.015                                 | 0.109                                                     | -0.017                                 | 0.086                                                     | -0.004                                 | 0.081                                                     | 0.014                                  | -0.018                                                    | 0.013                                  | -0.018                                                    |
| Kemeny–Young     | 10  | 0.110         | 0.203                  | 0.156        | 0.254                  | -0.068                                 | 0.151                                                     | -0.080                                 | 0.166                                                     | -0.035                                 | 0.113                                                     | -0.021                                 | 0.106                                                     | -0.018                                 | 0.089                                                     | -0.008                                 | 0.081                                                     | -0.001                                 | -0.023                                                    | -0.006                                 | -0.025                                                    |
| RRF              | 10  | 0.110         | 0.204                  | 0.157        | 0.255                  | -0.070                                 | 0.152                                                     | -0.085                                 | 0.167                                                     | -0.035                                 | 0.114                                                     | -0.022                                 | 0.107                                                     | -0.020                                 | 0.089                                                     | -0.012                                 | 0.081                                                     | -0.003                                 | -0.021                                                    | -0.008                                 | -0.024                                                    |
| Schulze          | 10  | 0.110         | 0.205                  | 0.155        | 0.256                  | -0.069                                 | 0.152                                                     | -0.080                                 | 0.169                                                     | -0.033                                 | 0.115                                                     | -0.021                                 | 0.108                                                     | -0.018                                 | 0.089                                                     | -0.008                                 | 0.082                                                     | -0.005                                 | -0.021                                                    | -0.008                                 | -0.022                                                    |

Table 12: NOISE-K results for **DermaMNIST** with ViT-B/16 (Full-matrix protocol), using the Kendall-based rank-noise geometry. The  $q$  column gives the selected explanation-prefix length. All metrics use  $p = 16$ ,  $k = 20$ , dataset-mean filling,  $\sigma_g = 0.15$ ,  $\pi_p = 0.05$ ,  $\sigma_s = 0.15$ , and  $\epsilon = 2/255$ . Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

| Method           | $q$ | Clean quality |                        |              |                        | Gaussian ( $g$ )                       |                                                           |                                        |                                                           | Salt-and-pepper ( $p$ )                |                                                           |                                        |                                                           | Speckle ( $s$ )                        |                                                           |                                        |                                                           | Adversarial ( $a$ )                    |                                                           |                                        |                                                           |
|------------------|-----|---------------|------------------------|--------------|------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|
|                  |     | F $\uparrow$  | $\bar{F}$ $\downarrow$ | C $\uparrow$ | $\bar{C}$ $\downarrow$ | R <sub>F</sub> <sup>g</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>g</sup> $\uparrow$ | R <sub>C</sub> <sup>g</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>g</sup> $\uparrow$ | R <sub>F</sub> <sup>p</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>p</sup> $\uparrow$ | R <sub>C</sub> <sup>p</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>p</sup> $\uparrow$ | R <sub>F</sub> <sup>s</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>s</sup> $\uparrow$ | R <sub>C</sub> <sup>s</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>s</sup> $\uparrow$ | R <sub>F</sub> <sup>a</sup> $\uparrow$ | R <sub><math>\bar{F}</math></sub> <sup>a</sup> $\uparrow$ | R <sub>C</sub> <sup>a</sup> $\uparrow$ | R <sub><math>\bar{C}</math></sub> <sup>a</sup> $\uparrow$ |
| Simple Averaging | 5   | 0.111         | 0.203                  | 0.156        | 0.254                  | -0.072                                 | 0.151                                                     | -0.084                                 | 0.167                                                     | -0.040                                 | 0.113                                                     | -0.031                                 | 0.106                                                     | -0.023                                 | 0.087                                                     | -0.012                                 | 0.079                                                     | 0.001                                  | -0.025                                                    | -0.001                                 | -0.025                                                    |
| Borda            | 5   | 0.099         | 0.206                  | 0.143        | 0.259                  | -0.062                                 | 0.153                                                     | -0.074                                 | 0.171                                                     | -0.024                                 | 0.115                                                     | -0.014                                 | 0.110                                                     | -0.018                                 | 0.088                                                     | -0.005                                 | 0.081                                                     | 0.027                                  | -0.021                                                    | 0.030                                  | -0.020                                                    |
| Kemeny–Young     | 5   | 0.110         | 0.206                  | 0.158        | 0.258                  | -0.071                                 | 0.153                                                     | -0.084                                 | 0.171                                                     | -0.036                                 | 0.115                                                     | -0.025                                 | 0.110                                                     | -0.020                                 | 0.089                                                     | -0.012                                 | 0.084                                                     | 0.013                                  | -0.018                                                    | 0.005                                  | -0.021                                                    |
| RRF              | 5   | 0.106         | 0.208                  | 0.157        | 0.259                  | -0.066                                 | 0.155                                                     | -0.082                                 | 0.172                                                     | -0.031                                 | 0.117                                                     | -0.027                                 | 0.111                                                     | -0.014                                 | 0.090                                                     | -0.010                                 | 0.085                                                     | 0.017                                  | -0.016                                                    | 0.010                                  | -0.016                                                    |
| Schulze          | 5   | 0.108         | 0.201                  | 0.156        | 0.252                  | -0.066                                 | 0.148                                                     | -0.082                                 | 0.165                                                     | -0.032                                 | 0.110                                                     | -0.026                                 | 0.104                                                     | -0.016                                 | 0.086                                                     | -0.008                                 | 0.080                                                     | 0.018                                  | -0.021                                                    | 0.017                                  | -0.024                                                    |